

AI USAGE DISCLOSURE

Student information

Name: Saliq Neyaz
Matriculation number: 4756033
Project title: Confidence, not conflict: which value→RT mapping best describes probability discounting, and does RT help?
Group members, if applicable: Ashhad Raza Quadri, Saliq Neyaz, Maria Alejandra Pabon Galindo

AI use statement

Please tick one option:

☐ **No AI tools used.**

I confirm that I did not personally use generative AI tools, AI-based writing assistants, translation tools, image generators, code assistants, or similar AI systems in preparing my contribution to this project.

☒ **AI tools used and disclosed below.**

I confirm that I used AI tools only in limited and transparent ways that preserved the independent character of my work. I remain fully responsible for the accuracy, academic integrity, citations, methodology, code, analyses, and final submitted content.

Details of AI use

If no AI tools were used, write “Not applicable” in the first row.

AI tool used	Purpose of use	Parts of the project affected	Extent of use / student responsibility
Claude Code (Anthropic 5.1)	Assistance with writing and debugging Python code: drafting implementations of routines I had specified, and diagnosing errors (optimiser convergence failures, numerical problems in the shifted log-normal log-likelihood).	Inference code for my components (Sec. 2.3): the MAP objective, the L-BFGS-B wrapper with multiple restarts, the MLE cross-check and the likelihood-curvature diagnostic. Reliability analysis code (Sec. 3.2, Table 3): Pearson/Spearman/ICC computation and the bootstrap CI routines. Associated plotting code for Figs. 2 and 5.	AI produced initial drafts of these routines from specifications I wrote; the choice of model, estimator, priors, diagnostics and statistics was mine. I read, corrected and tested all generated code, verified its output against the model-free estimates (Sec. 2.1) and the MLE cross-check, and can explain every line. No AI was used to interpret results or to decide what the analyses show.
Claude Code (Anthropic 5.1)	Language and grammar correction and rephrasing of text I had already drafted myself.	My own paragraphs in Sec. 2.3 (inference), Sec. 3.1–3.2 (identifiability and reliability) and the corresponding passages of Sec. 4.	Wording and clarity only. Every claim, number, argument and interpretation is my own; no passage was accepted without my review, and no substantive prose was generated from scratch by AI.

Prompt log

Did you retain a prompt log or notes documenting the AI-assisted steps?

- ☒ Yes, attached or available upon request.
☐ No AI tools were used.
☐ No prompt log was retained. Explanation: [explanation]

Individual contribution and responsibility

Briefly describe your own contribution to the project, including report writing, coding, modelling, analyses, figures, or interpretation of results:

I owned **RT model M2** (decision uncertainty, $\mu = a - b|2P - 1|$): its specification, psychological motivation, fitting and evaluation, including the result that it gives the best run B RT predictive log-likelihood (-1.492 nats per trial) and is the only model ranking that holds both in sample on run A and out of sample on run B. I also built the **inference machinery** (MAP objective, priors and their justification in Table 1, the L-BFGS-B optimiser with multiple restarts, the MLE cross-check and the likelihood-curvature identifiability diagnostic, both in Fig. 2) and the **reliability analysis** of Sec. 3.2, Fig. 5 and Table 3 (Pearson/Spearman/ICC, bootstrap CIs, paired CIs on reliability differences), which the report reads as generalisation across decision contexts rather than classical test–retest. I drafted the report text for Sec. 2.3, Secs. 3.1–3.2 and the corresponding passages of Sec. 4.

The work was divided equally in three. Shared components – the Rachlin hyperboloid valuation model and logistic choice rule, the shifted log-normal RT distribution, the model-comparison framework, the figures and this report – were developed jointly, and I reviewed and accept responsibility for the parts I contributed to.

In the case of a group project, I confirm that I have reviewed the shared report, code, analyses, and figures to which I contributed. I accept responsibility for my own contribution and for any AI-assisted material included in shared components that I reviewed, used, or submitted as part of the project.

Final confirmation

I confirm that I have not submitted AI-generated content as my own without review, verification, and disclosure. I have checked all factual claims, references, data, code, analyses, and interpretations myself. I have not entered sensitive, personal, confidential, or copyright-protected material into AI systems in violation of data protection, copyright, or course rules.

Name and signature: Saliq Neyaz

Date: 30 August 2026